

National University of Technology

Department of Artificial Intelligence



Assignment 01

RESEARCH PROBLEM SELECTION AND LITERATURE
EXPLORATION

Student Name: Aleena Tahir

Student ID: F23607001

Batch: AI-23

Instructor: Ms. Sumera Aslam

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1 Group Members

Sr.	Name	Roll Number
1	Aleena Tahir	F23607001

Table 1: Group members for the semester project.

2 Selected Research Problem

Project Title

Agent-Based Modeling of Adaptive Water Extraction Strategies Using Q-Learning Under Climate Stress Scenarios

This project models a shared water resource (river or aquifer) used by multiple irrigator agents. Instead of assigning fixed behavioral rules, each irrigator is equipped with a **Q-learning** algorithm that allows it to *learn* extraction strategies through trial and error. The simulation then tests whether sustainable resource management can emerge from decentralized learning, and how climate stress (drought scenarios) disrupts or reshapes these learned strategies.

The topic falls under the **Environment and Sustainability** domain (specifically *water resource sharing* and *overuse of common resources*) and introduces an AI/ML dimension through reinforcement learning agents.

3 Background and Importance of the Problem

Water scarcity is one of the most critical global challenges of the 21st century. Across the world, agricultural communities depend on shared freshwater sources such as rivers, canals, and underground aquifers for irrigation. When multiple farmers draw from the same source, each faces a fundamental dilemma: extracting more water increases their own crop yield in the short term, but if everyone extracts aggressively, the resource depletes and everyone suffers. This dilemma was famously described by Garrett Hardin in 1968 as the **Tragedy of the Commons**, where individually rational decisions lead to collectively disastrous outcomes.

The problem becomes even more urgent under climate change. Shifting rainfall patterns, prolonged droughts, and increasing temperatures reduce natural water regeneration rates. Regions that previously sustained stable irrigation systems now face sudden scarcity shocks. According to the United Nations World Water Development Report, nearly half of the global population will live in areas of high water stress by 2030. In Pakistan specifically, the Indus

River Basin, which supports over 90% of the country's agriculture, is classified as one of the most water-stressed basins in the world. Understanding how farming communities adapt their water usage behavior under these changing conditions is therefore not just an academic exercise but a matter of food security and economic survival.

What makes this problem particularly interesting to study is that real farmers do not follow fixed strategies. They observe their neighbors, learn from past seasons, and adapt. Traditional models of the commons problem assume either perfectly rational agents (game theory) or fixed behavioral types (cooperate/defect). Neither captures the *learning* process through which real communities develop, or fail to develop, sustainable water management norms. This gap motivates the use of reinforcement learning agents within an agent-based model: agents that discover strategies through experience rather than having them prescribed.

4 Why ABM is Suitable for This Problem

Agent-Based Modeling is uniquely suited for this problem for the following reasons:

4.1 Who Are the Agents?

The primary agents are **irrigator farmers** who share a common water resource. Each farmer independently decides how much water to extract each season. An optional secondary agent type is a **monitoring authority** that detects and penalizes over-extraction with some probability.

4.2 How Do They Interact?

Agents interact *indirectly* through the shared resource: one farmer's extraction reduces the water available for others. They also interact through *social observation*: each farmer can observe the extraction behavior of their neighbors (linked via a social network) and factor this into their decisions. In the reinforcement learning variant, agents' learned Q-values encode the history of these interactions.

4.3 How Does System Behavior Emerge?

The key emergent phenomena are:

- ▷ **Cooperation or collapse:** Whether the resource is sustained or depleted is not programmed; it emerges from the collective learning dynamics of all agents.
- ▷ **Oscillatory dynamics:** Agents may learn cooperation during good climate periods, then over-extract during droughts, causing cycles of depletion and recovery.

- ▷ **Inequality:** Some agents may learn exploitative strategies at the expense of others, producing unequal payoff distributions measurable via the Gini coefficient.

4.4 Why Not Other Approaches?

Table 2: Comparison of modeling approaches for the commons problem.

Approach	Limitation
Differential equations	Assume homogeneous, well-mixed agents; cannot model individual learning or spatial variation.
Classical game theory	Assume full rationality and known payoffs; real farmers learn under uncertainty.
System dynamics	Cannot represent agent heterogeneity, local interactions, or individual adaptation.
Network models	Capture topology but not resource dynamics or temporal learning processes.

Only ABM captures all of these simultaneously: heterogeneous agents learning through experience, spatial resource dynamics, social observation networks, and emergent collective outcomes.

4.5 Why NetLogo?

NetLogo provides built-in support for agent breeds, spatial patches, link-based social networks, and the BehaviorSpace tool for automated parameter sweeps. Its Models Library includes a Tragedy of the Commons model as a direct starting point. The Q-Learning Extension and Python Extension enable reinforcement learning integration without leaving the NetLogo environment.

5 Literature Review

5.1 Paper 1

Title	A Multi-Agent Reinforcement Learning Model of Common-Pool Resource Appropriation
Authors	Perolat, J., Leibo, J.Z., Zambaldi, V., Beattie, C., Tuyls, K., & Graepel, T.
Year	2017
Source	NeurIPS 2017 (found via Google Scholar)

Summary: This paper studied how independently learning deep RL agents behave in a common-pool resource game modeled as a partially observed Markov game on a 2D grid. Agents could harvest “apples” from a shared resource whose respawn rate depended on the remaining local stock. The researchers found that when the resource was scarce, agents learned aggressive appropriation and even used a “tagging” punishment mechanism to exclude rivals. With more abundant resources, cooperative harvesting emerged naturally. The study demonstrated that trial-and-error learning can reproduce many of the behavioral patterns observed in human commons experiments, including the relationship between exclusion, sustainability, and inequality. This paper is directly relevant because it establishes the feasibility of using RL agents to study commons dilemmas, though it used deep RL in a custom grid environment rather than tabular Q-learning in a water-specific ABM.

5.2 Paper 2

Title	An Agent-Based Conflict Resolution Model for Urban Water Resources Management
Authors	Darbandsari, P., Kerachian, R., Malakpour-Estalaki, S., & Khorasani, H.
Year	2020
Source	Sustainable Cities and Society, Vol. 57, 102112 (Elsevier)

Summary: This paper presented a novel conflict resolution framework for urban water management combining agent-based modeling with game theory. The model represented water stakeholders (municipal users, agricultural users, and environmental agencies) as agents in a leader-follower (Stackelberg) game structure. The framework was applied to a semi-urban-semi-agricultural case study in the Tehran region to evaluate management policies considering economic, social, and environmental dimensions. The study found that hierarchical game structures with a regulatory leader agent could steer stakeholders toward more equitable and sustainable water allocation. This paper is relevant because it demonstrates ABM

applied specifically to water resource conflicts among heterogeneous stakeholders, though it used fixed game-theoretic strategies rather than adaptive learning agents.

5.3 Paper 3

Title	Agent-Based Modelling of a Coupled Water Demand and Supply System at the Catchment Scale
Authors	Huber, L., Bahro, N., Leitinger, G., Tappeiner, U., & Strasser, U.
Year	2019
Source	Sustainability, Vol. 11(21), 6178, MDPI (found via Google Scholar)

Summary: This paper introduced *Aqua.MORE* (Agent-Based Modelling of Resources), a NetLogo-based platform for simulating coupled human–water systems at the catchment scale. The model represented both socioeconomic actors (water users and managers) and the water resource itself as agents, capturing how human demand decisions and natural supply dynamics interact over time. Applied to an Alpine catchment, the model evaluated scenarios of changing water demand under different management policies. The study showed that even modest shifts in user behavior could significantly affect catchment-level water availability. This paper is directly relevant because it provides a validated example of water resource ABM implemented in NetLogo, including the resource regeneration dynamics and stakeholder heterogeneity that this project will build upon, although it did not incorporate any learning or adaptive behavior for its agents.

5.4 Paper 4

Title	An Oscillating Tragedy of the Commons in Replicator Dynamics with Game-Environment Feedback
Authors	Weitz, J.S., Eksin, C., Paarporn, K., Brown, S.P., & Ratcliff, W.C.
Year	2016
Source	PNAS, 113(47), E7518–E7525 (found via Google Scholar)

Summary: This paper proposed a coevolutionary game theory framework where the environment and agent strategies evolve together. Applied to the Tragedy of the Commons, the model showed that overuse of a resource changes the environment, which in turn changes the incentive structure, leading to *oscillatory* dynamics: the system cycles between depleted and

replete resource states and between cooperative and defective behavior. The authors identified conditions under which the tragedy can be averted (when cooperation is favored in the depleted state) and showed that the outcome depends on the strength of coupling between strategies and environment. This paper is relevant because it provides the theoretical foundation for the oscillatory cooperation-collapse dynamics that this project expects to observe, particularly under climate shock scenarios where environmental feedback drives behavioral shifts.

6 Comparison of Existing Studies

Table 3: Comparative analysis of reviewed literature.

Aspect	Perolat et al.	Darbandsari et al.	Huber et al.	Weitz et al.
Agent Type	Deep RL agents (A3C)	Game-theoretic stakeholders	Water users & managers	Evolutionary populations
Learning	Yes (deep RL)	No (fixed strategies)	No (rule-based)	Replicator dynamics
Resource	Abstract apples	Urban water allocation	Catchment-level water	Abstract commons
Climate Stress	Not modeled	Not modeled	Scenario-based	Environment feedback
Tool	Custom grid engine	MATLAB / custom	NetLogo	Mathematical (ODEs)
Key Finding	Scarcity drives aggression; abundance enables cooperation	Hierarchical governance improves equity	Behavioral shifts affect catchment water	Oscillatory cooperation-defection cycles

Common themes: All four studies recognize that individual agent behavior drives collective resource outcomes, and that the structure of interactions (social, hierarchical, or environmental) critically shapes whether cooperation or tragedy prevails.

Key differences: The studies diverge fundamentally on agent cognition. Perolat et al. use sophisticated deep RL, while Darbandsari et al. and Huber et al. use fixed or rule-based behavior, and Weitz et al. use evolutionary dynamics. None combine *tabular Q-learning agents* with *explicit climate stress scenarios* in a *water-specific* ABM, which is precisely the gap this project targets.

Lessons for this project: From Perolat et al., the insight that scarcity promotes aggression while abundance enables cooperation will inform hypothesis generation. From Darband-

sari et al., the importance of modeling heterogeneous stakeholder types. From Huber et al., the NetLogo-based architecture for coupled human–water systems. From Weitz et al., the oscillatory dynamics framework that predicts how climate shocks may destabilize learned cooperation.

7 Research Gap and Extension Idea

Most existing agent-based models of water commons dilemmas assign *fixed behavioral types* (selfish, cooperative, tit-for-tat) to agents at initialization. The agents do not learn from experience or adapt to changing conditions. The few studies that incorporate learning, such as Perolat et al. (2017), use deep reinforcement learning in abstract, non-water-specific grid environments that do not model realistic resource regeneration dynamics or climate variability.

Meanwhile, climate change is making water scarcity increasingly stochastic and unpredictable, rendering fixed-strategy models inadequate. Farmers in real irrigation systems learn, observe their neighbors, and adjust. Yet **no published ABM applies Q-learning agents to a water commons dilemma under climate stress scenarios**. There is no existing model that answers: can decentralized RL agents discover sustainable water extraction strategies on their own? And what happens to those learned strategies when a drought hits?

This project fills that gap by equipping each irrigator agent with an independent tabular Q-learning algorithm, placing them in a shared water resource with logistic regeneration dynamics, and systematically testing their learned behavior under stable, gradually declining, shock-based, and stochastic climate scenarios. The use of simple tabular Q-learning (rather than deep RL) also makes the model transparent, interpretable, and implementable within NetLogo, an additional practical contribution.

8 Research Question

Primary Research Question:

How does the discount factor (γ) in Q-learning agents affect the emergence of sustainable water extraction strategies under varying climate stress scenarios in an agent-based commons model?

Sub-questions:

1. Can Q-learning agents discover cooperative extraction strategies without external enforcement mechanisms?
2. How do learned strategies break down under sudden climate shocks (drought events), and how quickly do agents re-adapt?

3. At what monitoring intensity (probability of detecting and penalizing over-extraction) do enforcement mechanisms become unnecessary because agents have learned self-regulation?

9 Preliminary Model Components

9.1 Agents

- ▷ **Irrigator farmers** (primary): Each farmer has a Q-table, extraction amount, payoff, and social observation capabilities. Three behavioral variants will be tested: Q-learning agents, fixed-strategy agents (always-cooperate, always-defect, tit-for-tat), and mixed populations.
- ▷ **Monitor agents** (optional): Represent a water authority that detects over-extraction with some probability and imposes fines.

9.2 Environment

- ▷ A shared **water resource** (river or aquifer) modeled as a global variable with **logistic regeneration**: $W(t+1) = W(t) + r \cdot c_f(t) \cdot W(t) \cdot \left(1 - \frac{W(t)}{K}\right) - \sum_i e_i(t)$, where r is the regeneration rate, K is carrying capacity, $c_f(t)$ is the climate factor, and $\sum_i e_i(t)$ is total extraction.
- ▷ **Spatial patches**: Blue patches represent the river (color intensity proportional to water level), green patches represent farmland.
- ▷ **Social network**: Links between farmers represent observation connections.

9.3 Agent Behaviors

- ▷ **Observe**: Read water level, own past harvest, and neighbors' average extraction.
- ▷ **Decide**: Select extraction level using ϵ -greedy Q-learning (5 discrete actions from zero to maximum extraction).
- ▷ **Extract**: Remove water from the shared resource.
- ▷ **Learn**: Update Q-table using the update rule: $Q(s, a) \leftarrow Q(s, a) + \alpha [R + \gamma \max_{a'} Q(s', a') - Q(s, a)]$.
- ▷ **Adapt**: Decay exploration rate ϵ over time to shift from exploration to exploitation.

9.4 Possible Parameters (NetLogo Sliders)

Parameter	Symbol	Range
Number of farmers	N	5 – 100
Learning rate	α	0.01 – 0.5
Discount factor	γ	0.5 – 0.99
Exploration rate (initial)	ϵ_0	0.1 – 1.0
Climate scenario		Stable / Gradual / Shock / Stochastic
Monitoring probability	p_{detect}	0 – 1.0
Regeneration rate	r	0.01 – 0.3
Proportion of Q-learners		0% – 100%

Table 4: Key parameters to be controlled via NetLogo interface sliders.

10 Conclusion

This assignment has established the foundation for a semester-long research project on agent-based modeling of adaptive water extraction strategies. The selected problem, the Tragedy of the Commons in shared water resources, is a globally critical issue that becomes increasingly urgent under climate change. ABM is uniquely suitable because it captures the heterogeneous, adaptive, and locally interacting nature of real farming communities, which equation-based and game-theoretic models cannot represent.

The literature review identified a clear research gap: while RL agents have been applied to abstract commons games (Perolat et al., 2017) and ABMs have modeled water conflicts with fixed strategies (Darbandsari et al., 2020; Huber et al., 2019), no existing model combines Q-learning agents with water-specific resource dynamics under climate stress. The research question, how the discount factor in Q-learning agents affects the emergence of sustainable extraction under climate variability, is clear, focused, and testable through NetLogo simulation. The preliminary model components (agents, environment, behaviors, and parameters) provide a ready blueprint for the next assignment where implementation will begin.

11 References

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